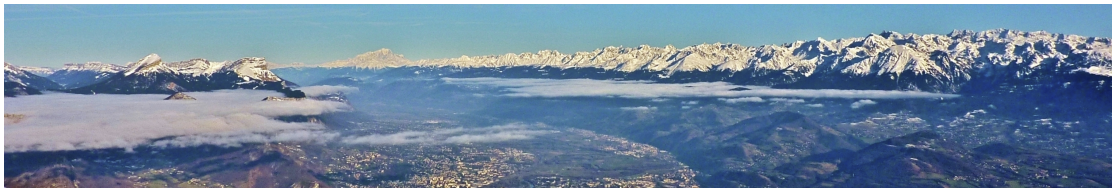


Physics-Informed Machine Learning at the Atomic Scale

Martin Uhrín

Computational Atomistic Methods & Machine Learning, SIMaP, UGA





Materials informatics



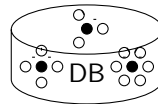
- High-throughput quantum mechanical calculations
- Turnkey workflows for calculation of properties
- Large quantities of readily reusable property calculations

Invariant and equivariant machine learning



- Theory of atomistic structure representations
- Inversion of invariant fingerprints
- Highly data-efficient equivariant neural networks

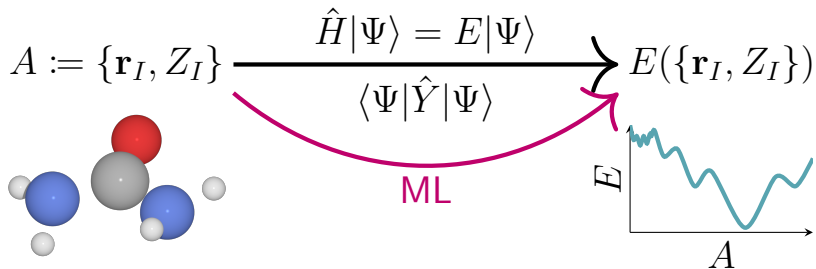
Generative models



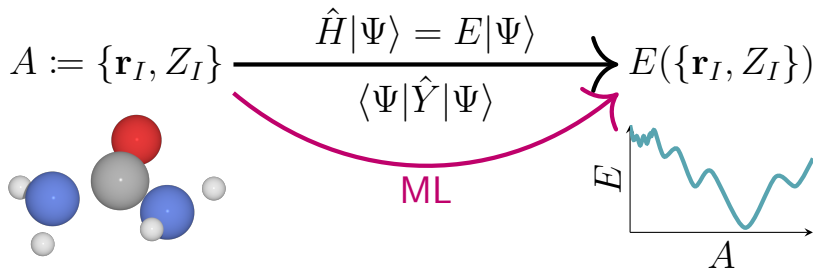
- Linking 3D geometry to properties
- Motif based construction of molecules and materials
- Coupling experiment and theory

Physics inspired machine learning models

Accelerating property prediction with machine learning

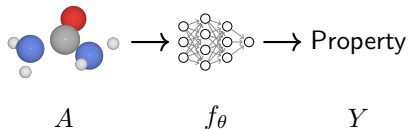


Accelerating property prediction with machine learning

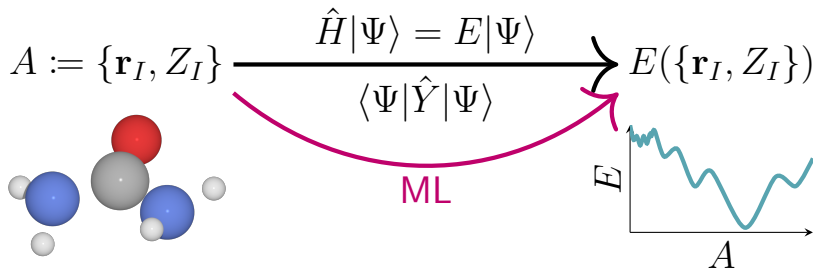


Our ML models learns to map from atomic coordinates to properties:

$$f_{\theta} : A \rightarrow Y,$$

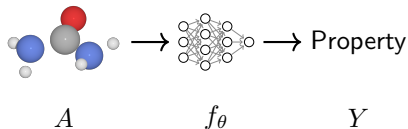


Accelerating property prediction with machine learning

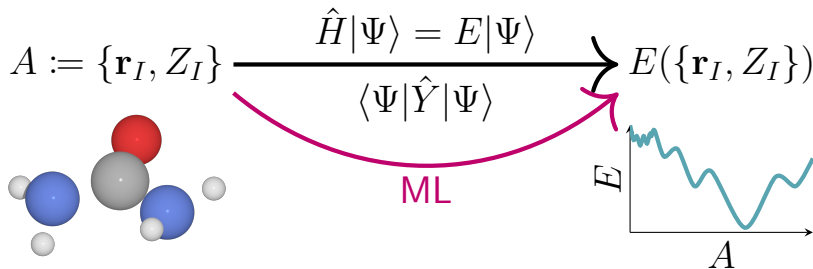


Our ML models learn to map from atomic coordinates to properties:

$$f_{\theta} : A \rightarrow Y,$$

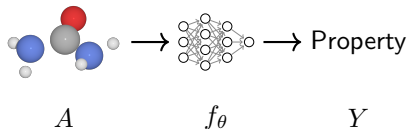


Accelerating property prediction with machine learning



Our ML models learn to map from atomic coordinates to properties:

$$f_{\theta} : A \rightarrow Y,$$

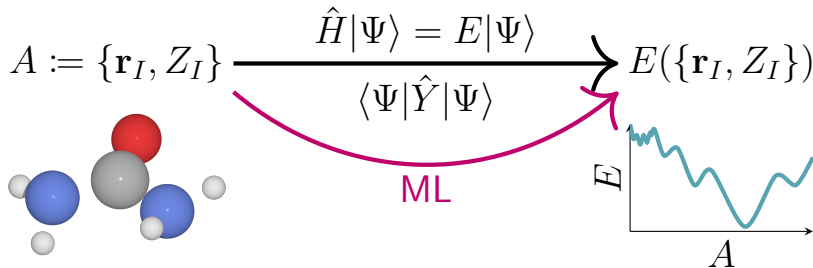


but A is not a suitable input.

We want something that respects the symmetries of the underlying physical laws:

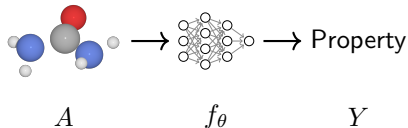
- permutation
- translation
- rotation

Accelerating property prediction with machine learning



Our ML models learn to map from atomic coordinates to properties:

$$f_{\theta} : A \rightarrow Y,$$



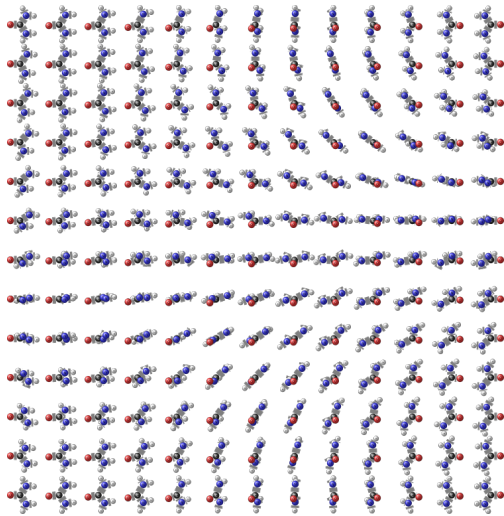
but A is not a suitable input. We want something that respects the symmetries of the underlying physical laws:

- permutation
- translation
- rotation

furthermore, for generative models we want representations that are:

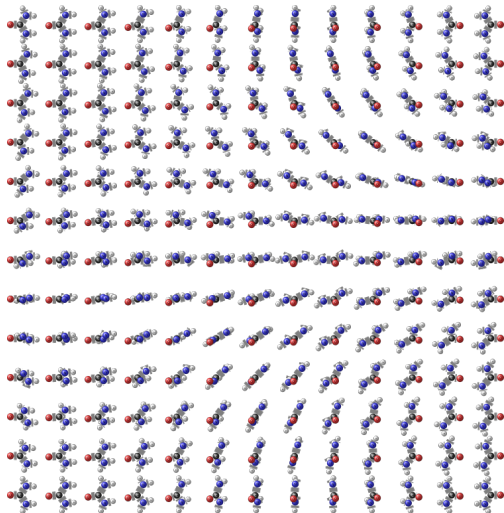
- smooth,
- continuous
- complete

Learning with no symmetry awareness



requires hundreds fold augmentation for $O(3)$

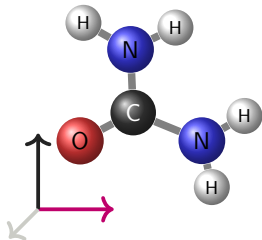
Learning with no symmetry awareness



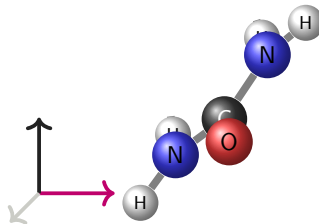
requires hundreds fold augmentation for $O(3)$

Learning with symmetry-awareness





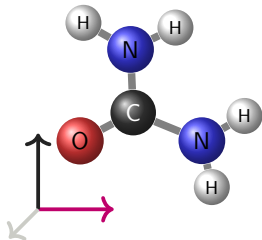
a)



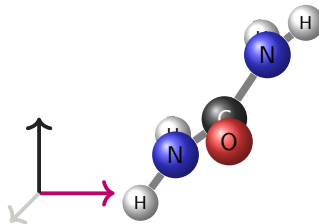
b)

Regular learning model will see
a) and b) as completely
different inputs

All functions



a)



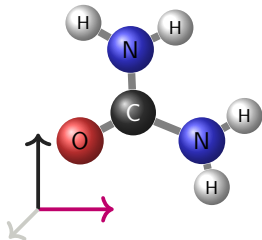
b)

Regular learning model will see
a) and b) as completely
different inputs

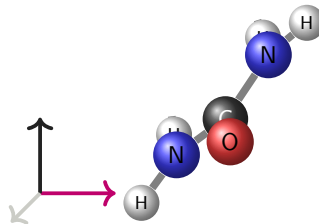
Invariant model will see the
same inputs.

All functions

Invariant



a)



b)

Regular learning model will see
a) and b) as completely
different inputs

Invariant model will see the
same inputs.

Equivariant model will see
the same inputs, and carry
around gemetric (non-scalar)
information.

All functions

Invariant

Equivariant

Lifting the hood on equivariant neural networks

Invariant model

$$f_{\theta}(D_X[g]x) = f_{\theta}(x)$$

for the group g .

Lifting the hood on equivariant neural networks

Invariant model

$$f_{\theta}(D_X[g]x) = f_{\theta}(x)$$

for the group g . **Equivariant model:**

$$D_Y[g]f_{\theta}(x) = f_{\theta}(D_X[g]x) \quad \forall g \in G, \forall x \in X$$

Natural way of extending ML models to handle non-scalar inputs and outputs (vectors, tensors),

Lifting the hood on equivariant neural networks

Invariant model

$$f_{\theta}(D_X[g]x) = f_{\theta}(x)$$

for the group g . **Equivariant model:**

$$D_Y[g]f_{\theta}(x) = f_{\theta}(D_X[g]x) \quad \forall g \in G, \forall x \in X$$

Natural way of extending ML models to handle non-scalar inputs and outputs (vectors, tensors),

Fully-connected layer

$$\phi \left(\begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ x_2 \\ y_2 \\ z_2 \end{bmatrix}^{\top} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \end{bmatrix} \dots \right) = [x_3 \dots]$$

$\phi(x)$ - activation function

Lifting the hood on equivariant neural networks

Equivariant layer

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \otimes \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1x_2 & x_1y_2 & x_1z_2 \\ y_1x_2 & y_1y_2 & y_1z_2 \\ z_1x_2 & z_1y_2 & z_1z_2 \end{bmatrix}$$

Irreducible form:

 $l = 0$ scalar - $w_1(x_1x_2 + y_1y_2 + z_1z_2)$
 $l = 1$ vector - $w_2 \begin{bmatrix} y_1z_2 - z_1x_2 \\ z_1x_2 - x_1z_2 \\ x_1y_2 - x_2y_1 \end{bmatrix}$
 $l = 2$ vector - $w_3 \begin{bmatrix} x_1z_2 + z_1x_2 \\ x_1y_2 + y_1x_2 \\ 2y_1y_2 - x_1x_2 - z_1z_2 \\ y_1z_2 - z_1y_2 \\ z_1z_2 - x_1x_2 \end{bmatrix}$

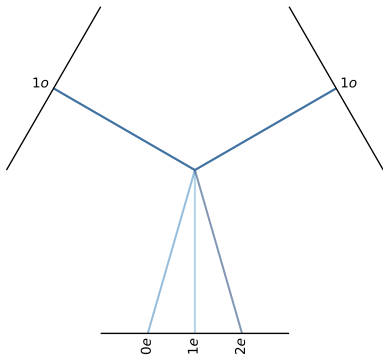
Fully-connected layer

$$\phi \left(\begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ x_2 \\ y_2 \\ z_2 \end{bmatrix}^\top \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \end{bmatrix} \dots \right) = [x_3 \dots]$$

 $\phi(x)$ - activation function

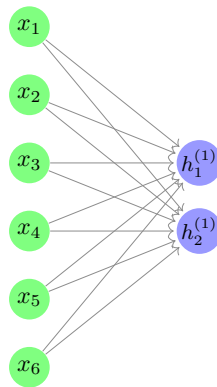
Lifting the hood on equivariant neural networks

Equivariant layer



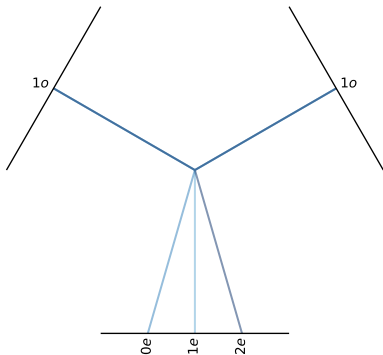
Outputs: $i \times 0e \oplus j \times 1e \oplus k \times 2e$

Fully-connected layer



Lifting the hood on equivariant neural networks

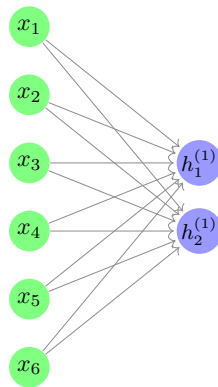
Equivariant layer



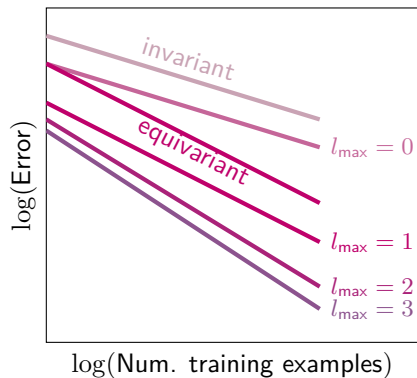
Outputs: $i \times 0e \oplus j \times 1e \oplus k \times 2e$

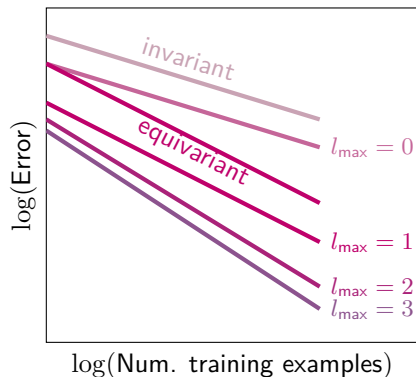
- Maintains symmetry (cannot *rotate* input)
- Minimal amount of parameters

Fully-connected layer



- No constraint on transformation (can scale/rotate/translate inputs)
- Many parameters





System		NequIP ^{a)}	NequIP ^{b)}	NequIP ^{c)}	DeepMD
Liquid Water	Energy	-	1.6	1.7	1.0
	Forces	11.9	49.4	11.6	40.4
Ice Ih (b)	Energy	-	2.5	4.3	0.7
	Forces	10.2	55.8	9.9	43.3
Ice Ih (c)	Energy	-	3.9	10.2	0.7
	Forces	12.0	27.7	11.7	26.8
Ice Ih (d)	Energy	-	2.6	12.7	0.8
	Forces	9.8	23.2	9.5	25.4

NequIP model trained on $< 0.1\%$ of DeepMD model

S. Batzner et al., *Nature Communications* **13**, 2453 (2022)

python ecosystem for development of equivariant learning

Complete ecosystem built in JAX and e3nn-jax:

tensorial

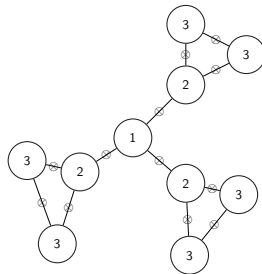
- Native support for equivariant graph convolutions
- Complete flexibility over input and output types:
 - per-node, per-edge or global
 - arbitrary node attributes (scalars, tensors)
- Implemented in **JAX: 4-20x speedup over pytorch**

reax

- pytorch lightning-like model training

e3md

- Built in implementations of
 - NequIP
 - Allegro
 - MACE



```
def f(x):
    return ** 2

# Evaluate gradient
jax.grad(f)(5.)

# Multiple inputs, scalar output
jax.grad(dft_energy(atom_positions))

# Multiple inputs, multiple outputs
jax.jacfwd(get_bands(atom_positions))
```

Electric and magnetic response

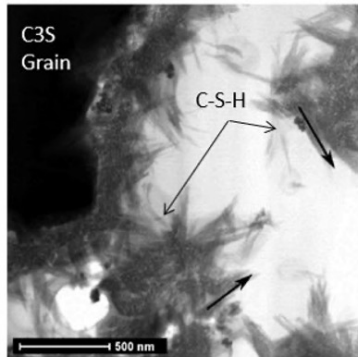


Laura Mismetti
IBM



Ziga Casar
Princeton

- Main product of the hydration process of cement 50% hardened cement paste
- Its nucleation and growth are the controlling steps of the early hydration reaction
- Forms a nanoporous network which controls the ion transport (Cl^-) and consequently influences the service-life of concrete structures

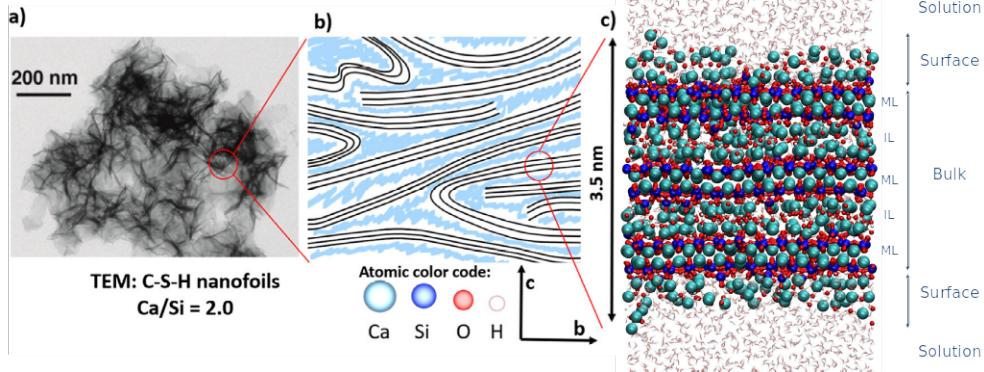


CSH micrograph

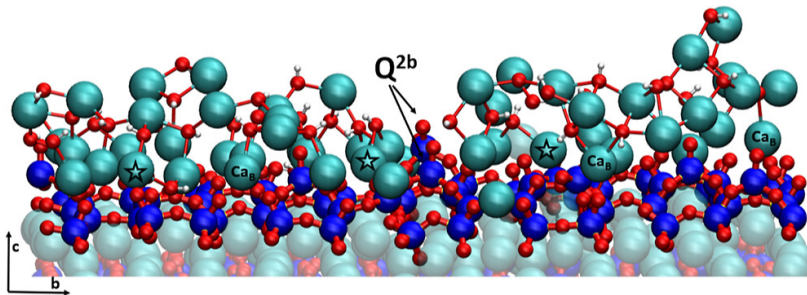
Nanocrystalline
structure

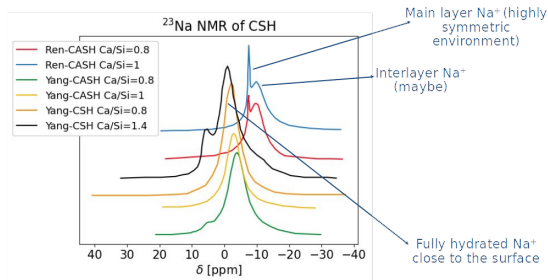
Sheets of
thickness 3.5nm

Layered structure



- Ions in solution (Na^+ , Cl^- , ...)
- Interactions at the interface could affect cohesion and long-term durability of cement-based materials
- High Na^+ concentration: it seems it removes Ca^{2+} and OH^- , lowering the surface charge
- Nuclear Magnetic Resonance (NMR) to get info about the local environment
- Insights on Na^+ incorporation sites





- Nuclear Magnetic Resonance (NMR) to get info about the local environment
- Sodium incorporation sites not yet understood

Goal: Deepen the understanding of sodium incorporation in CSH exploiting NMR data (simulations + experiments)



Mattia Ragni
UGA

$$E = U - \vec{\mu} \cdot \vec{B}$$

$\vec{\mu}$ - magnetic dipole moment

$\vec{B}$ - external magnetic field



Mattia Ragni
UGA

$$E = U - \vec{\mu} \cdot \vec{B}$$

$\vec{\mu}$ - magnetic dipole moment

$\vec{B}$ - external magnetic field

NMR shielding tensor

$$\sigma_{\alpha\beta}^I = \frac{\partial^2 E}{\partial B_\alpha \partial \mu_\beta^I} = \vec{B} \otimes \vec{\mu}^I$$



Mattia Ragni
UGA

NMR shielding tensor

$$\sigma_{\alpha\beta}^I = \frac{\partial^2 E}{\partial B_{\alpha} \partial \mu_{\beta}^I} = \vec{B} \otimes \vec{\mu}^I$$

rank 2 non-symmetric tensor, irreps $0e \oplus 1e \oplus 2e$

$$E = U - \vec{\mu} \cdot \vec{B}$$

$\vec{\mu}$ - magnetic dipole moment

$\vec{B}$ - external magnetic field

$$\sigma_{\alpha\beta} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}$$



Mattia Ragni
UGA

$$E = U - \vec{\mu} \cdot \vec{B}$$

$\vec{\mu}$ - magnetic dipole moment

$\vec{B}$ - external magnetic field

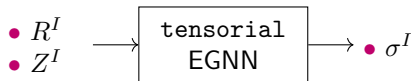
NMR shielding tensor

$$\sigma_{\alpha\beta}^I = \frac{\partial^2 E}{\partial B_\alpha \partial \mu_\beta^I} = \vec{B} \otimes \vec{\mu}^I$$

$$\sigma_{\alpha\beta} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}$$

rank 2 non-symmetric tensor, irreps $0e \oplus 1e \oplus 2e$

Model - GNN:

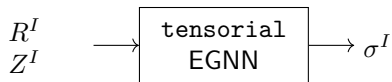


Injecting more physics

NMR shielding tensor

$$\sigma_{\alpha\beta}^I = \frac{\partial^2 E}{\partial B_\alpha \partial \mu_\beta^I} = \vec{B} \otimes \vec{\mu}^I$$

rank 2 non-symmetric tensor, irreps $0e \oplus 1e \oplus 2e$

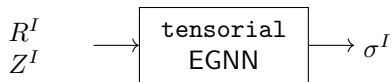


Injecting more physics

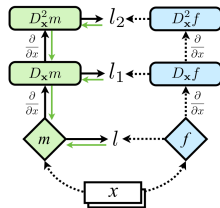
NMR shielding tensor

$$\sigma_{\alpha\beta}^I = \frac{\partial^2 E}{\partial B_\alpha \partial \mu_\beta^I} = \vec{B} \otimes \vec{\mu}^I$$

rank 2 non-symmetric tensor, irreps $0e \oplus 1e \oplus 2e$



Sobolev learning

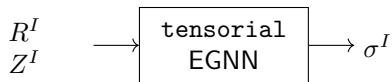


Injecting more physics

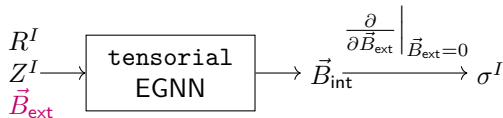
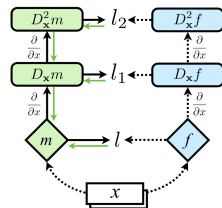
NMR shielding tensor

$$\sigma_{\alpha\beta}^I = \frac{\partial^2 E}{\partial B_\alpha \partial \mu_\beta^I} = \vec{B} \otimes \vec{\mu}^I$$

rank 2 non-symmetric tensor, irreps $0e \oplus 1e \oplus 2e$



Sobolev learning

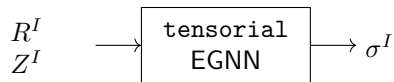


Injecting more physics

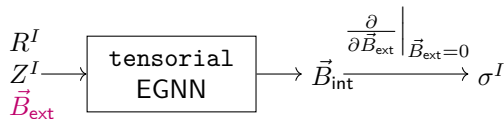
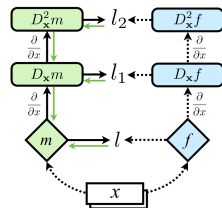
NMR shielding tensor

$$\sigma_{\alpha\beta}^I = \frac{\partial^2 E}{\partial B_\alpha \partial \mu_\beta^I} = \vec{B} \otimes \vec{\mu}^I$$

rank 2 non-symmetric tensor, irreps $0e \oplus 1e \oplus 2e$



Sobolev learning



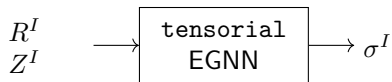
All functions

Equivariant

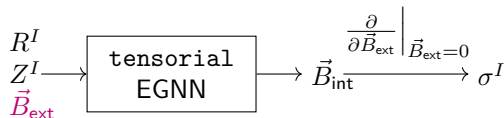
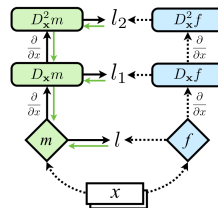
NMR shielding tensor

$$\sigma_{\alpha\beta}^I = \frac{\partial^2 E}{\partial B_\alpha \partial \mu_\beta^I} = \vec{B} \otimes \vec{\mu}^I$$

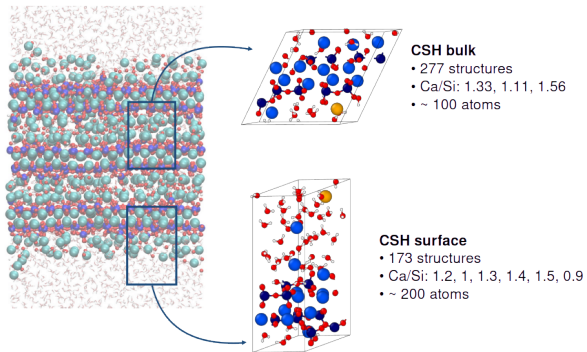
rank 2 non-symmetric tensor, irreps $0e \oplus 1e \oplus 2e$



Sobolev learning



All functions Conservative Equivariant



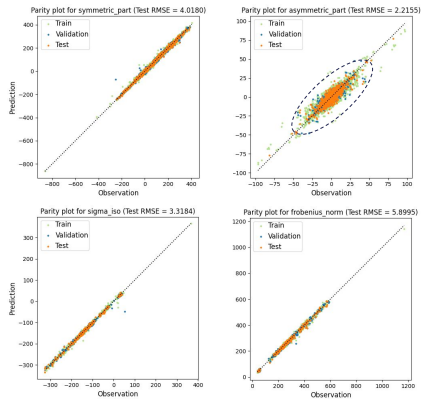
Bulk and surface CSH structures with water

- Representative CSH bulk and surface (containing Na⁺) generated with the brick model
- Pre-relaxation with empirical FF + DFT relaxation
- QE+GIPAW method to calculate NMR tensor (3x3) for every atom

500 small organic molecules from the QM9 dataset, each containing up to 9 heavy atoms (C, N, O, and F)

EGNN + Differentiation learning

Train RMSE: 4.35 ppm
Val RMSE: 5.27 ppm
Test RMSE: 4.59 ppm



Enforcing Curl-Freeness via Automatic Differentiation

Rank 0 $\rightarrow$ Rank 1

Latent Scalar Field:

$$E(\mathbf{x}) \in \mathbb{R} \quad (\text{Total Energy})$$

Observed Vector Field:

$$\mathbf{F}(\mathbf{x}) = -\nabla E(\mathbf{x}) = - \begin{bmatrix} \frac{\partial E}{\partial x} \\ \frac{\partial E}{\partial y} \\ \frac{\partial E}{\partial z} \end{bmatrix}$$

Mathematical Identity:

$$\nabla \times \mathbf{F} = \nabla \times (-\nabla E) \equiv \mathbf{0}$$

Clairaut's Theorem (C^2 Smoothness):

$$\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = -\frac{\partial^2 E}{\partial x \partial y} + \frac{\partial^2 E}{\partial y \partial x} = 0$$

Enforcing Curl-Freeness via Automatic Differentiation

Rank 0 $\rightarrow$ Rank 1

Latent Scalar Field:

$$E(\mathbf{x}) \in \mathbb{R} \quad (\text{Total Energy})$$

Observed Vector Field:

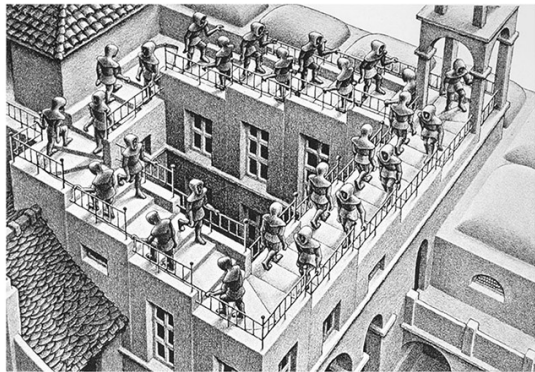
$$\mathbf{F}(\mathbf{x}) = -\nabla E(\mathbf{x}) = - \begin{bmatrix} \frac{\partial E}{\partial x} \\ \frac{\partial E}{\partial y} \\ \frac{\partial E}{\partial z} \end{bmatrix}$$

Mathematical Identity:

$$\nabla \times \mathbf{F} = \nabla \times (-\nabla E) \equiv \mathbf{0}$$

Clairaut's Theorem (C^2 Smoothness):

$$\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = -\frac{\partial^2 E}{\partial x \partial y} + \frac{\partial^2 E}{\partial y \partial x} = 0$$



Characterization in Reverse: Structure Resolution via Differentiable ML

The Paradigm Shift

Instead of using ML solely for forward prediction (Structure $\rightarrow$ Spectrum), end-to-end differentiable models allow us to **invert the characterization process** via gradient descent.

Characterization in Reverse: Structure Resolution via Differentiable ML

The Paradigm Shift

Instead of using ML solely for forward prediction (Structure $\rightarrow$ Spectrum), end-to-end differentiable models allow us to **invert the characterization process** via gradient descent.

1. The General Framework

- 1 Start with a candidate atomic structure ($\mathbf{R}$).
- 2 Predict physical observables ($\hat{\mathbf{O}}$) using physics-informed ML model.
- 3 Define a loss against experimental spectra ($\mathbf{O}_{\text{exp}}$):

$$\mathcal{L} = \|\hat{\mathbf{O}}(\mathbf{R}) - \mathbf{O}_{\text{exp}}\|^2$$

- 4 **Backpropagate** to update atomic positions:

$$\mathbf{R}_{t+1} = \mathbf{R}_t - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{R}}$$

Characterization in Reverse: Structure Resolution via Differentiable ML

The Paradigm Shift

Instead of using ML solely for forward prediction (Structure $\rightarrow$ Spectrum), end-to-end differentiable models allow us to **invert the characterization process** via gradient descent.

1. The General Framework

- 1 Start with a candidate atomic structure ($\mathbf{R}$).
- 2 Predict physical observables ($\hat{\mathbf{O}}$) using physics-informed ML model.
- 3 Define a loss against experimental spectra ($\mathbf{O}_{\text{exp}}$):

$$\mathcal{L} = \|\hat{\mathbf{O}}(\mathbf{R}) - \mathbf{O}_{\text{exp}}\|^2$$

- 4 **Backpropagate** to update atomic positions:

$$\mathbf{R}_{t+1} = \mathbf{R}_t - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{R}}$$

2. Proof-of-Concept: NMR Relaxation

- **Setup:** Take a relaxed ground-state structure and compute its reference NMR tensors (σ_{ref}).
- **Perturbation:** Intentionally displace a single ion away from its true equilibrium position.
- **Inversion:** Minimize the tensor discrepancy using automatic differentiation through the ML engine:

$$\min_{\mathbf{R}_{\text{ion}}} \|\hat{\sigma}(\mathbf{R}) - \sigma_{\text{ref}}\|^2$$

Resolving structure from predictions using gradients

$$E = U - \vec{P} \cdot \vec{\epsilon}$$

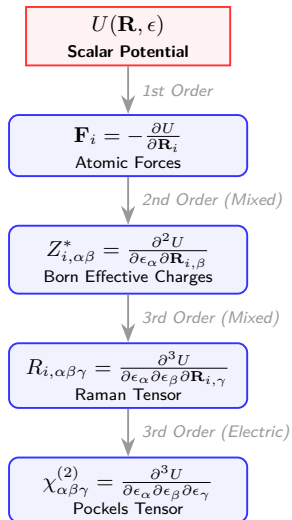
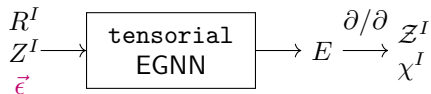
Born effective charge tensor

Raman tensor

$$Z_{\alpha\beta}^I = \frac{\partial^2 E}{\partial \epsilon_\alpha \partial \tau_\beta^I}$$

$$\chi_{\alpha\beta\gamma}^I = -\frac{1}{\Omega} \frac{\partial^3 E}{\partial \epsilon_\alpha \partial \epsilon_\beta \partial \tau_\gamma^I}$$

Model





Lorenzo Bastonero
Uni Bremen



Alessandro D'Urso
EPFL

$$E = U - \vec{P} \cdot \vec{\epsilon}$$

$\vec{P}$ - polarization

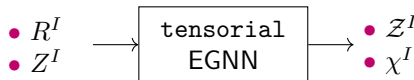
$\vec{\epsilon}$ - external electric field

Born effective charge tensor

$$Z_{\alpha\beta}^I = \frac{\partial^2 E}{\partial \epsilon_\alpha \partial \tau_\beta^I} = \vec{\epsilon} \otimes \vec{\tau}^I$$

rank 2 non-symmetric tensor, irreps $0e \oplus 1e \oplus 2e$

Model - double headed EGNN:



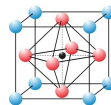
Raman tensor - Can be calculated via derivatives of the enthalpy:

$$\begin{aligned} \chi_{\alpha\beta\gamma}^I &= -\frac{1}{\Omega} \frac{\partial^3 E}{\partial \epsilon_\alpha \partial \epsilon_\beta \partial \tau_\gamma^I} \\ &= -\frac{1}{\Omega} \vec{\epsilon} \otimes \vec{\epsilon} \otimes \vec{\tau}^I \end{aligned}$$

rank 3 tensor symmetric in α, β irreps $2 \times 1o \oplus 2o \oplus 3o$

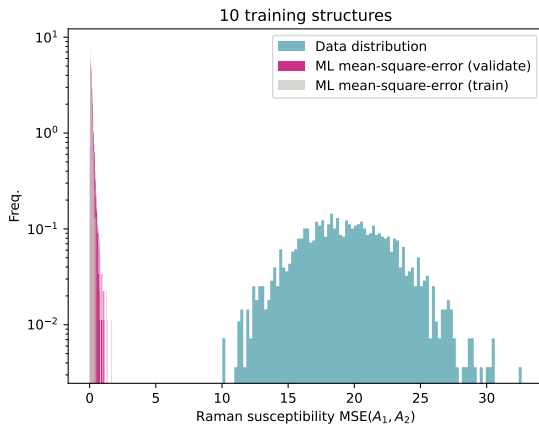
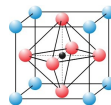
Electric response results

50 structures BaTiO_3 , 135 atoms, at 400K with random displacements from equilibrium.



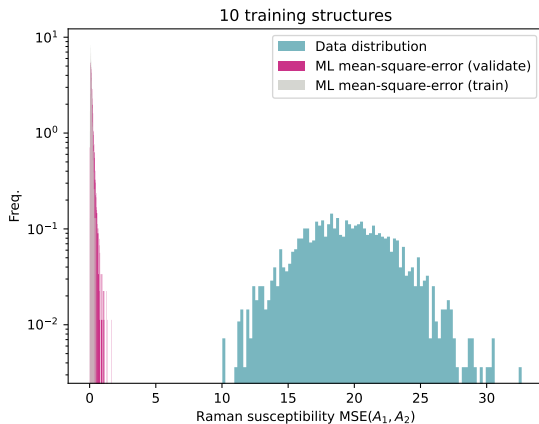
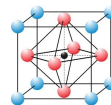
Electric response results

50 structures BaTiO_3 , 135 atoms, at 400K with random displacements from equilibrium.

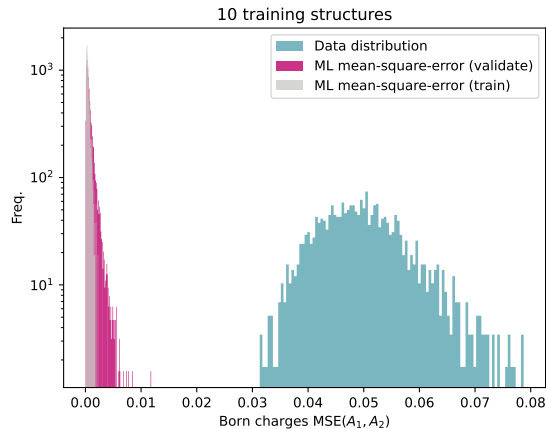


Raman susceptibilities

50 structures BaTiO_3 , 135 atoms, at 400K with random displacements from equilibrium.



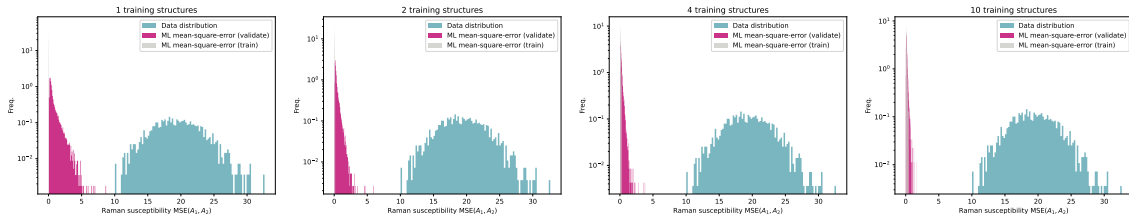
Raman susceptibilities



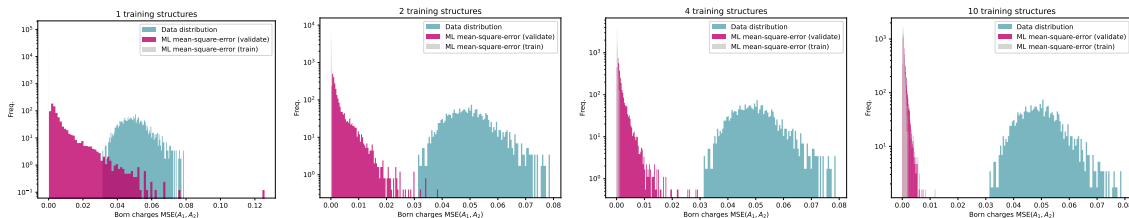
Born effective charges

Accuracy as function of training set size

Raman susceptibilities



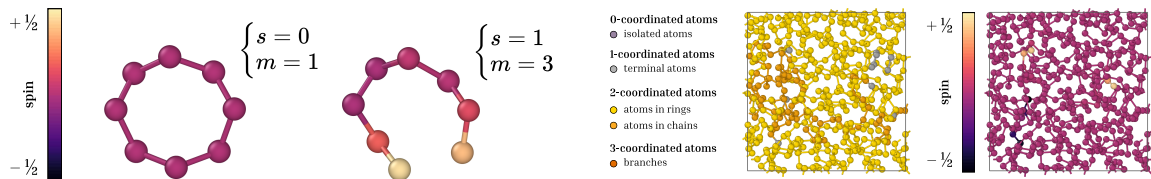
Born effective charges



The AI frontier: explicit electronic structure

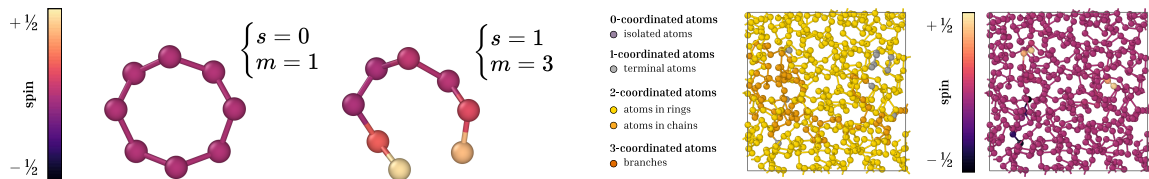
Collaboration with S. Salomoni, A. Salcedo and M. Saitta, ENS

The problem

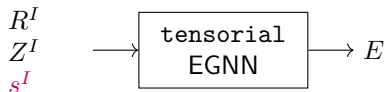


Collaboration with S. Salomoni, A. Salcedo and M. Saitta, ENS

The problem

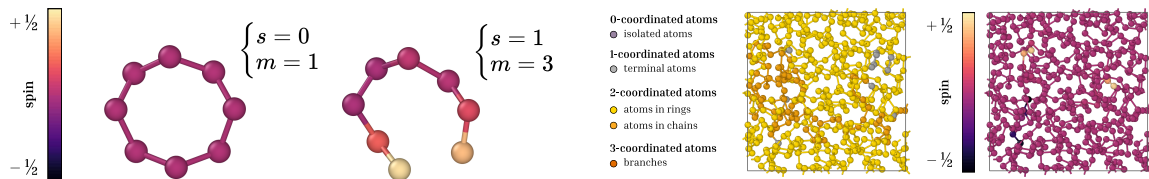


Model architecture

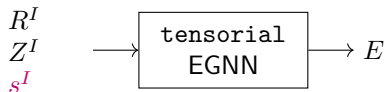


Collaboration with S. Salomoni, A. Salcedo and M. Saitta, ENS

The problem



Model architecture



But where do we get the spins, s^I ?

The Adiabatic Bridge: Inner-Loop Spin Optimization

Implicit Potential Surface

$$V_{\text{eff}}(\mathbf{R}) = \min_{\mathbf{S}} [E_{\text{ML}}(\mathbf{R}, \mathbf{S}) + \mathcal{P}(\mathbf{S})]$$

$$\mathcal{P}(\mathbf{S}) = \alpha \left\| \sum_i \mathbf{S}_i - S_{\text{T}} \right\|^2$$

Conservative Autodiff Forces

$$\mathbf{F}_I = -\nabla_{\mathbf{R}_I} V_{\text{eff}}(\mathbf{R})$$

The Adiabatic Bridge: Inner-Loop Spin Optimization

Implicit Potential Surface

$$V_{\text{eff}}(\mathbf{R}) = \min_{\mathbf{S}} [E_{\text{ML}}(\mathbf{R}, \mathbf{S}) + \mathcal{P}(\mathbf{S})]$$

$$\mathcal{P}(\mathbf{S}) = \alpha \left\| \sum_i \mathbf{S}_i - S_{\text{T}} \right\|^2$$

Conservative Autodiff Forces

$$\mathbf{F}_I = -\nabla_{\mathbf{R}_I} V_{\text{eff}}(\mathbf{R})$$

- **Adiabatic Separation:** Spins instantly relax to ground state ($T_{\text{elec}} = 0 \text{ K}$) relative to nuclei.
- **Enforced Compensation:** Global zero-net-moment state preserved without freezing local spin degrees of freedom.

The Adiabatic Bridge: Inner-Loop Spin Optimization

Implicit Potential Surface

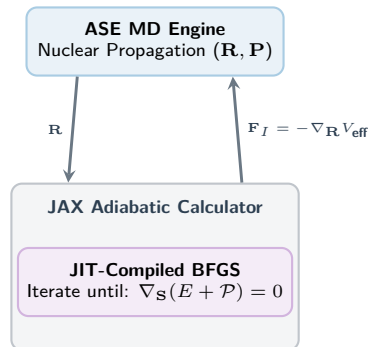
$$V_{\text{eff}}(\mathbf{R}) = \min_{\mathbf{S}} [E_{\text{ML}}(\mathbf{R}, \mathbf{S}) + \mathcal{P}(\mathbf{S})]$$

$$\mathcal{P}(\mathbf{S}) = \alpha \left\| \sum_i \mathbf{S}_i - S_{\text{T}} \right\|^2$$

Conservative Autodiff Forces

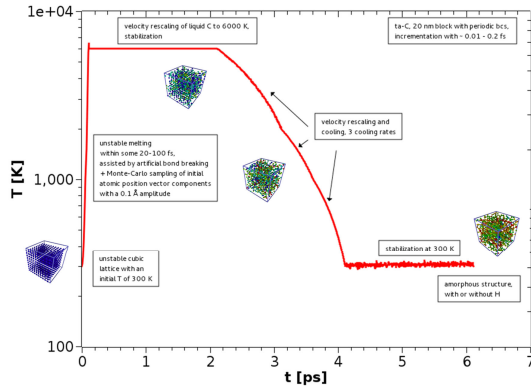
$$\mathbf{F}_I = -\nabla_{\mathbf{R}_I} V_{\text{eff}}(\mathbf{R})$$

- **Adiabatic Separation:** Spins instantly relax to ground state ($T_{\text{elec}} = 0 \text{ K}$) relative to nuclei.
- **Enforced Compensation:** Global zero-net-moment state preserved without freezing local spin degrees of freedom.



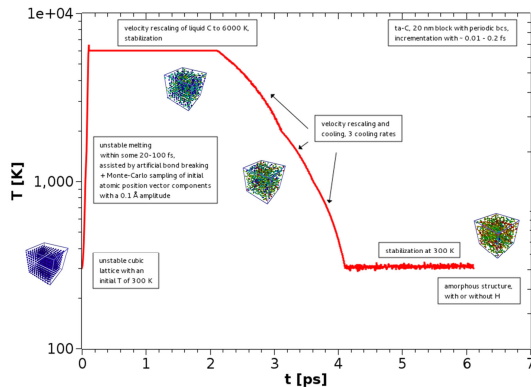
Generative models of atomic configurations

Example: generating atomic configurations in amorphous solids



Timo Hakala, Kenneth Holmberg and Anssi Laukkanen. Lubricants. 9. 30. (2021).

Example: generating atomic configurations in amorphous solids

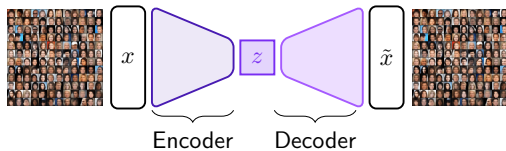


Given an example structure(s), can we teach a generative machine learning model to generate novel examples, bypassing the need for further molecular dynamics?

Timo Hakala, Kenneth Holmberg and Anssi Laukkanen. Lubricants. 9. 30. (2021).

The Variational Autoencoder

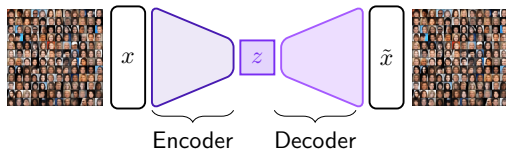
The autoencoder



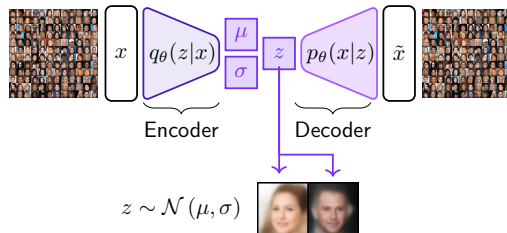
$$\mathcal{L} = (x - \tilde{x})^2$$

The Variational Autoencoder

The autoencoder

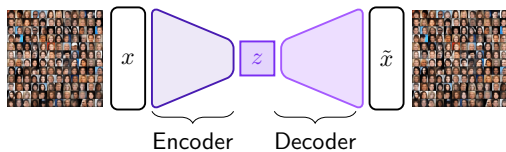


The *variational* autoencoder



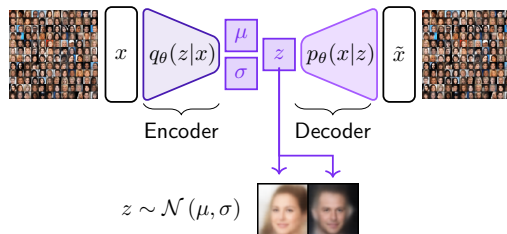
The Variational Autoencoder

The autoencoder



$$\mathcal{L} = (x - \tilde{x})^2$$

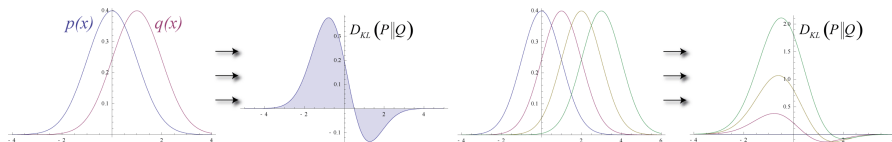
The *variational* autoencoder



$$\mathcal{L} = (x - \tilde{x})^2 + \sum_j KL(q_j(z|x)||p(z))$$

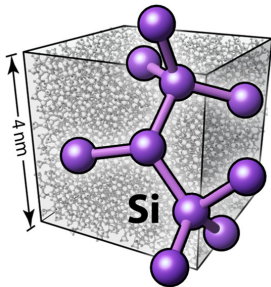
Kullback-Leibler divergence

$$D_{KL}(P \parallel Q) = \int_{-\infty}^{\infty} p(x) \log \left(\frac{p(x)}{q(x)} \right) dx$$



Symmetry-aware representation of local atomic environments

Direct representation



$$x = \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_N & y_N & z_N \end{bmatrix}$$

Not a suitable input to a learning model.

Consider

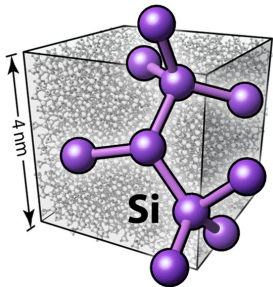
$$x' = xQ$$

where Q is a rotation matrix.

V. L. Deringer et al., Journal of Physical Chemistry Letters **9**, 2879–2885 (2018)

Symmetry-aware representation of local atomic environments

Direct representation



V. L. Deringer et al., *Journal of Physical Chemistry Letters* **9**, 2879–2885 (2018)

$$x = \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_N & y_N & z_N \end{bmatrix}$$

Not a suitable input to a learning model.

Consider

$$x' = xQ$$

where Q is a rotation matrix.

Symmetry-aware representation

$$G = \begin{bmatrix} x^1 \cdot x^1 & \dots & x^1 \cdot x^N \\ \vdots & \ddots & \vdots \\ x^N \cdot x^1 & \dots & x^N \cdot x^N \end{bmatrix} = \begin{bmatrix} \|x^1\|^2 & \dots & \|x^1\| \|x^N\| \cos \theta_{1N} \\ \vdots & \ddots & \vdots \\ \|x^N\| \|x^1\| \cos \theta_{N1} & \dots & \|x^N\|^2 \end{bmatrix}$$

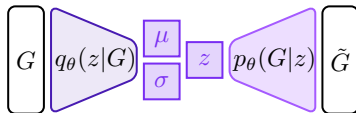
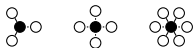
This representation is **rotationally invariant**.

$$G = xx^T = (xQ)(xQ)^T = xQQ^T x^T = xIx^T$$

Variational autoencoder for atomic motifs

Training

- ① For each atom in unit cell, extract local atomic environment up to r_{cut} . Keeps closest n atoms



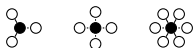
minimise

$$\mathcal{L} = (G - \tilde{G})^2 + \sum_j KL(q_j(z|G)||p(z))$$

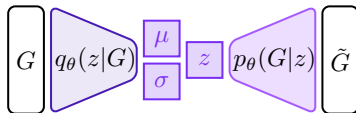
Variational autoencoder for atomic motifs

Training

- 1 For each atom in unit cell, extract local atomic environment up to r_{cut} . Keeps closest n atoms



- 2 Calculate Gram matrix $x^i \cdot x^j$, keep upper triangular part, $j \geq i$



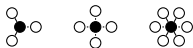
minimise

$$\mathcal{L} = (G - \tilde{G})^2 + \sum_j KL(q_j(z|G) || p(z))$$

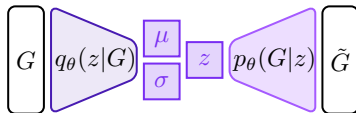
Variational autoencoder for atomic motifs

Training

- 1 For each atom in unit cell, extract local atomic environment up to r_{cut} . Keeps closest n atoms



- 2 Calculate Gram matrix $x^i \cdot x^j$, keep upper triangular part, $j \geq i$
- 3 Generate permutation copies of atom labels i (data augmentation) e.g. $[1, 2, 3], [1, 3, 2], [2, 1, 3]$, etc



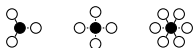
minimise

$$\mathcal{L} = (G - \tilde{G})^2 + \sum_j KL(q_j(z|G) || p(z))$$

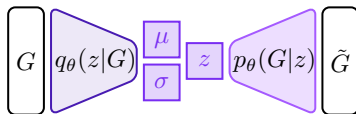
Variational autoencoder for atomic motifs

Training

- 1 For each atom in unit cell, extract local atomic environment up to r_{cut} . Keeps closest n atoms



- 2 Calculate Gram matrix $x^i \cdot x^j$, keep upper triangular part, $j \geq i$
- 3 Generate permutation copies of atom labels i (data augmentation) e.g. $[1, 2, 3]$, $[1, 3, 2]$, $[2, 1, 3]$, etc
- 4 Train VAE using gradient-based optimisation

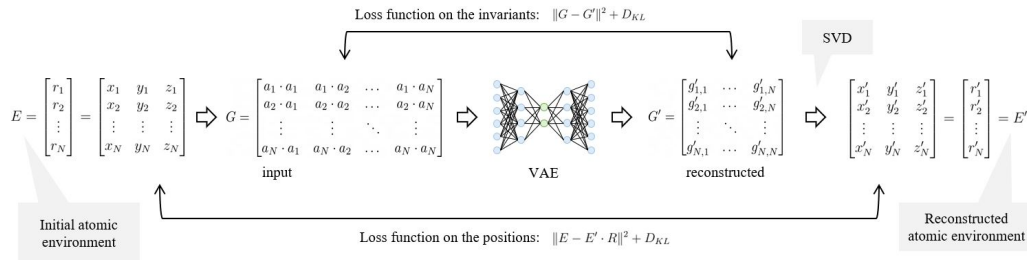


minimise

$$\mathcal{L} = (G - \tilde{G})^2 + \sum_j KL(q_j(z|G) || p(z))$$

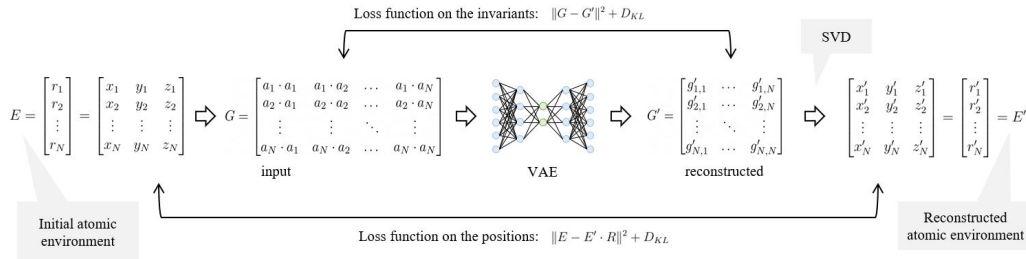
Training and generating

Training



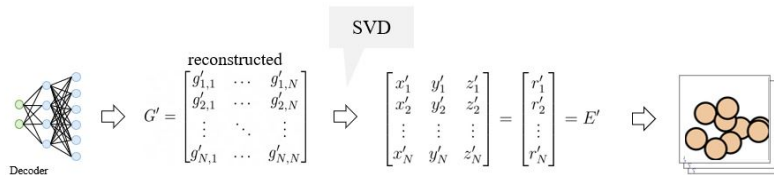
Training and generating

Training



Generating

Draw n_Z samples from $\mathcal{N}(0, 1)$



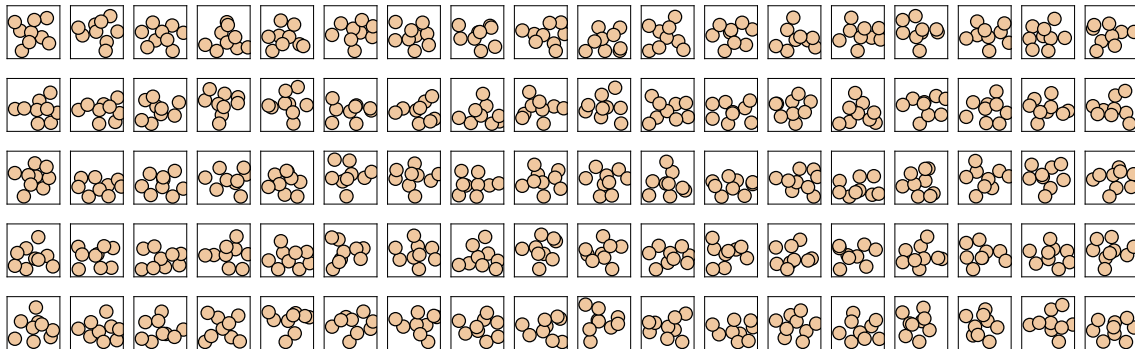
Variational autoencoder for atomic motifs

All tests performed in amorphous Si, 512 atom unit cells generated using ML potential trained on DFT¹.

- Radial cutoff: 4 Å
- 8 atoms per environment
- latent space: 8 neurons (compared to $3n - 6 = 18$ DoFs)
- Encoder architecture 36-28-18-8 with tanh activations

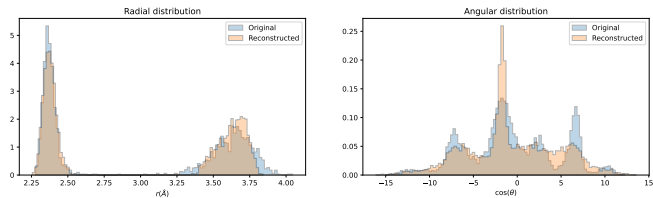
Data normalisation

- $G'_{ii} = (G_{ii} - \mu_{\text{diag}})/\sigma_{\text{diag}}$
- $G'_{ij} = (G_{ij} - \mu_{\text{off-diag}})/\sigma_{\text{off-diag}}, i < j$



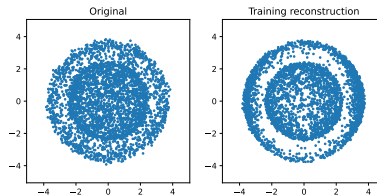
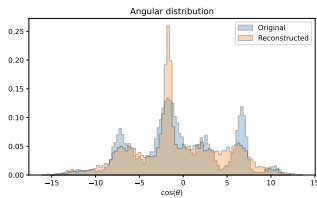
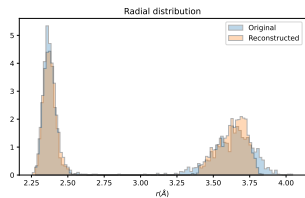
Model performance

Training reconstruction



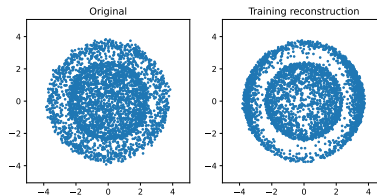
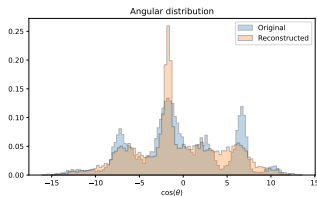
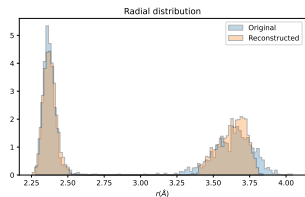
Model performance

Training reconstruction

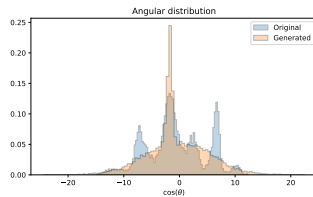
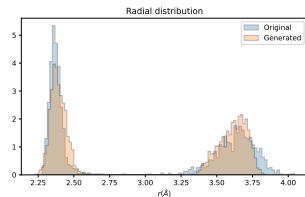


Model performance

Training reconstruction

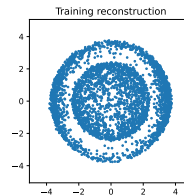
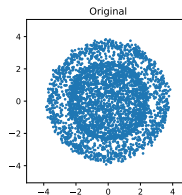
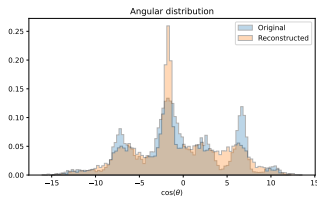
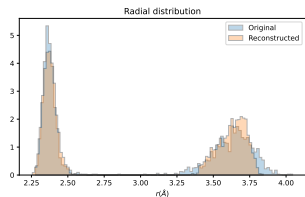


Generated samples

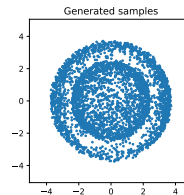
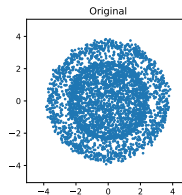
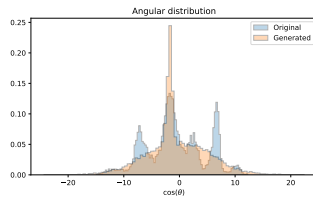
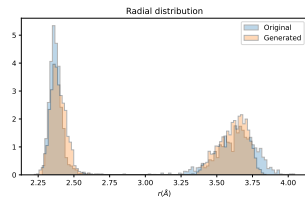


Model performance

Training reconstruction

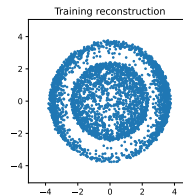
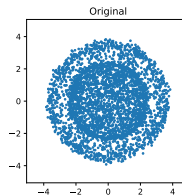
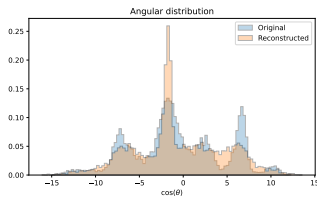
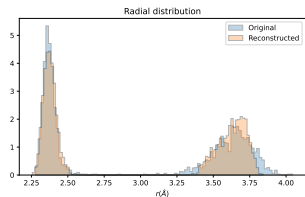


Generated samples

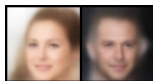
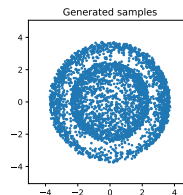
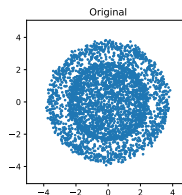
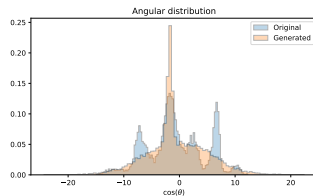
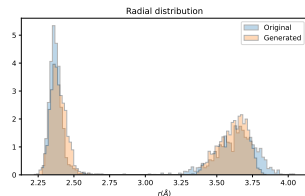


Model performance

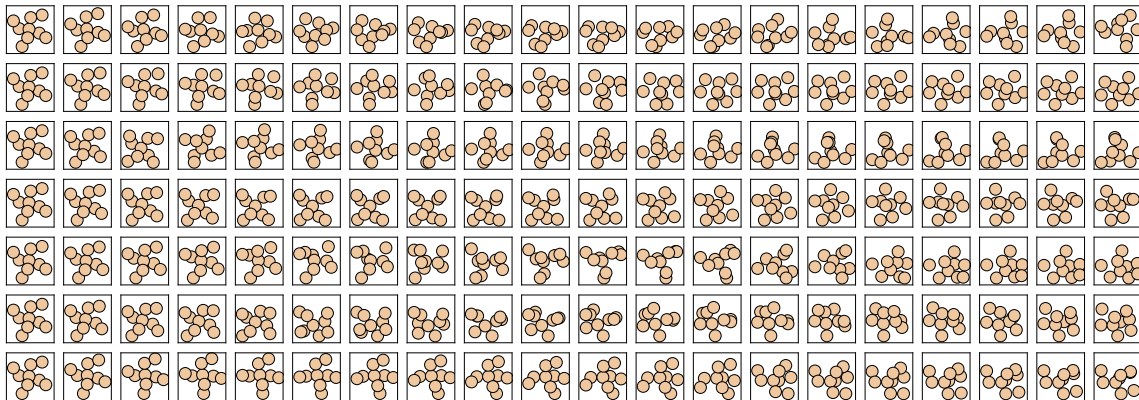
Training reconstruction



Generated samples



Interpolating between motifs



Interpolating between motifs

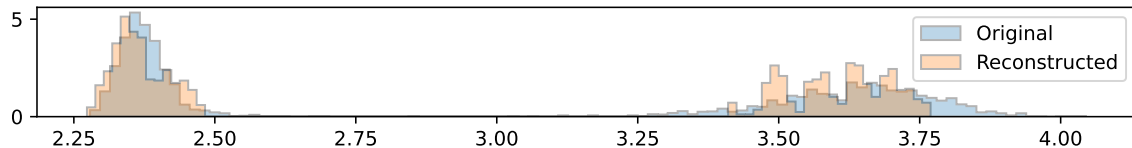
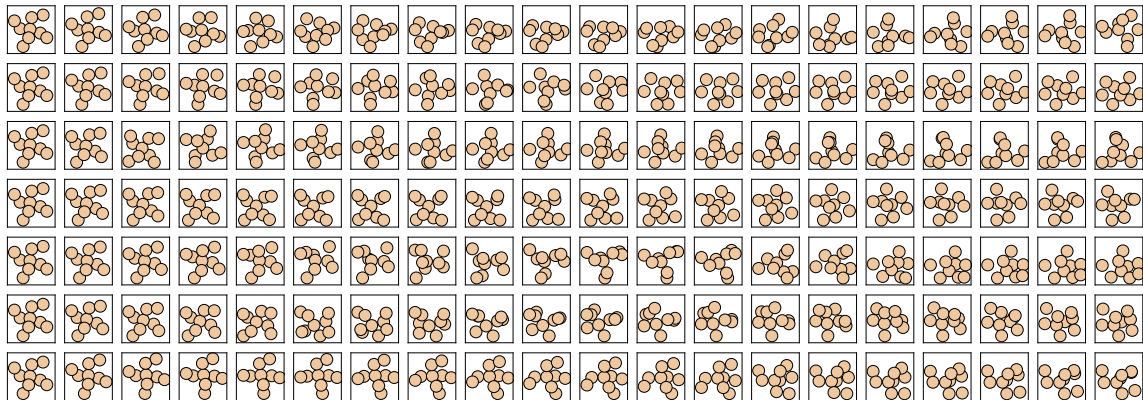


Image inpainting

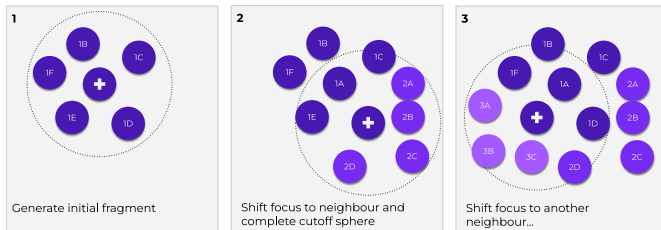


Atom infilling: Building complete unit cells

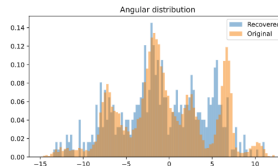
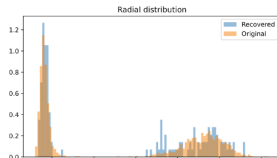
Image inpainting



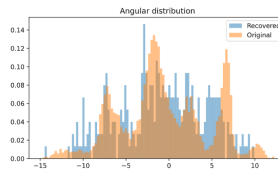
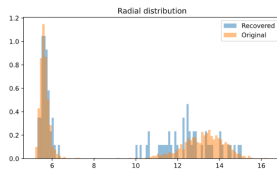
Environment infilling



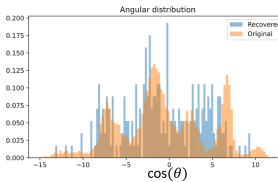
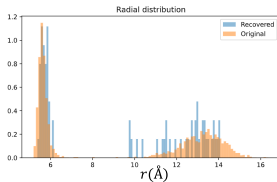
Fixed atoms: 7

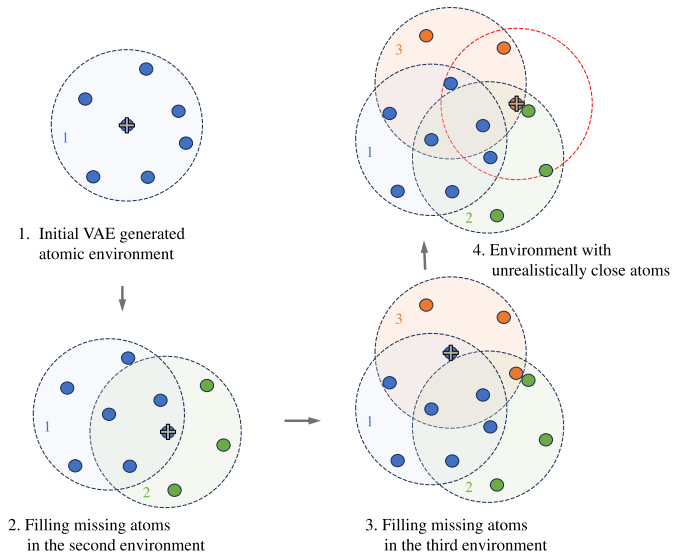


Fixed atoms: 6



Fixed atoms: 5





Summary

Euclidean neural networks naturally **encode the assumption that atomic systems exist in 3D Euclidean space**

This inductive bias can make them more **robust, data-efficient and accurate when predicting properties**

GNNs give us a **flexible way to express model inputs and outputs** and perform multi-target learning

Euclidean neural networks naturally **encode the assumption that atomic systems exist in 3D Euclidean space**

Where is this going?

Multiscale approaches to **bridge time and length scales** (coarse graining, etc)

This inductive bias can make them more **robust, data-efficient and accurate when predicting properties**

GNNs give us a **flexible way to express model inputs and outputs** and perform multi-target learning

Euclidean neural networks naturally **encode the assumption that atomic systems exist in 3D Euclidean space**

This inductive bias can make them more **robust, data-efficient and accurate when predicting properties**

GNNs give us a **flexible way to express model inputs and outputs** and perform multi-target learning

Where is this going?

Multiscale approaches to **bridge time and length scales** (coarse graining, etc)

Coupling to experiment i) using ML to help with interpretation, ii) using experiment to train ML models

Generative models that can **output atomic configurations with high probability of having the desired properties**

Summary
Acknowledgements



Nicola Marzari
EPFL



Tess Smidt
MIT



Mario Geiger
Nvidia



In the tutorial you will see how to predict electric field response properties using equivariant graph neural networks.

- Can run on cluster or on Google Colab (recommended - see links in github)

Backup

Accuracy as function of num. training iterations

